TENET SYSTEMS

Technical Architecture Whitepaper

Settlement & Liquidity Orchestration for Institutional Digital Assets

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Abstract

Tenet Systems is a settlement and liquidity orchestration layer for institutional digital assets. In its simplest form:

> **PRIMARY DEFINITION** *Tenet lets Financial Institutions trade and settle digital assets automatically from one pool of cash.*

Financial institutions submit a single structured trade intent. Tenet handles everything that follows — best price discovery across a network of institutional liquidity providers, orchestrated delivery and simultaneous settlement extraction from the FI's pre-funded vault, cross-chain settlement routing to LPs on their native chains, and a complete clearing ledger record of every order lifecycle event.

FIs never interact with LPs. LPs never interact with FIs. Tenet is the only counterparty either side ever sees.

This document describes the full technical architecture of Tenet Systems, including the four-engine model, the vault architecture, the Reservation Engine, the clearing ledger, the order execution lifecycle, multi-LP split logic, settlement routing, failure handling, the legal agreement framework, and the competitive landscape.

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1. The Problem

The infrastructure that traditional prime brokerage built over decades does not exist for digital assets. As financial institutions move to allocate capital to digital assets and tokenized real-world assets, they encounter four compounding structural problems. The first two are the most damaging.

1.1 Capital Inefficiency

Financial institutions entering digital asset markets must pre-fund multiple accounts across multiple chains simultaneously. Capital sits locked and idle in each of these accounts, waiting for orders that may not come for hours or days. An institution operating across Ethereum, Solana, Arbitrum, Base, and Polygon may have tens of millions of dollars frozen across five separate positions — none of which can cover the others.

This is dead capital. It earns nothing. It cannot be redeployed. It cannot be consolidated. Every new chain the institution wants to operate on requires another pre-funded account, compounding the inefficiency. The cost of this idle capital is a direct and recurring drag on institutional returns — and it is entirely a function of missing infrastructure, not market risk.

1.2 Operational Complexity

Operating across digital asset chains is not a single integration problem. It is a continuous, compounding operational burden. Institutions must monitor bridge reliability across dozens of chain pairs, manage fragmented liquidity provider relationships with different custody arrangements and settlement timelines, rebuild settlement infrastructure per chain pair as new chains emerge, and maintain internal engineering capacity to handle the failure modes of each.

None of this is alpha-generating activity. It is pure overhead — engineering time, compliance review, operational risk, and management attention that could be deployed elsewhere. The result is that most institutions either limit themselves to a handful of chains, accept the overhead as a cost of doing business, or stay out of digital assets entirely.

1.3 No Viable Settlement Infrastructure

Delivery versus payment — the simultaneous exchange of asset and cash that underlies all traditional securities settlement — does not exist at scale for digital assets. Between asset delivery and settlement extraction, institutions carry real counterparty exposure. There is no institutional-grade, chain-agnostic clearing layer that handles settlement automatically and guarantees LP payout on native chains. The infrastructure that DTCC provides for traditional markets simply has no equivalent for digital assets.

This is not a gap that existing players have filled. Custodians hold assets but do not route or settle. Exchanges match orders but leave settlement to the parties. Bridges move assets but introduce new risks. No single layer orchestrates the full flow from intent to settled LP payout.

1.4 Fragmented Liquidity

No single venue has sufficient depth across chains. An institution seeking to execute a large ETH order must identify, negotiate with, and operationally connect to multiple liquidity providers, each on different chains with different custody arrangements. There is no unified routing layer, no best-execution logic across LPs, and no single counterparty absorbing the complexity. The

institution bears all of it internally, and the result is worse pricing, slower execution, and higher operational cost than the market should require.

1.5 The Consequence

Institutional capital has not fully entered digital asset markets not because of lack of interest but because of lack of infrastructure. The capital inefficiency is too high. The operational burden is too great. The settlement layer does not exist. The liquidity is too fragmented. Tenet solves all four. One pool of cash. One intent. Everything else is handled.

2. What Tenet Is

PRIMARY DEFINITION *Tenet lets Financial Institutions trade and settle digital assets automatically from one pool of cash.*

TECHNICAL DEFINITION *Tenet Systems is the settlement and liquidity orchestration layer for institutional digital assets and tokenized real-world assets.*

INFRASTRUCTURE DEFINITION *Tenet is the clearing infrastructure digital asset markets have been waiting for — asset-light, network-driven, and positioned at the highest-margin layer of the institutional stack.*

The word orchestration is deliberate. Tenet does not take principal risk like a dealer. It does not match orders like an exchange. It does not move assets like a bridge. It orchestrates — coordinating every component of the institutional trade lifecycle from intent submission to LP settlement payout, without taking ownership of assets at any point in the process.

2.1 What Tenet Is Not

Category	Why Tenet Is Not This
A broker-dealer	Brokers take the other side of trades from their balance sheet. Tenet is asset-light. Co-signed vaults hold assets. Tenet never owns them.
An exchange	Exchanges match buyers and sellers directly. Tenet routes, clears, and settles end-to-end. FIs and LPs never interact.
A bridge	Bridges move assets cross-chain. Tenet delegates cross-chain routing to CCTP and Everclear. Bridging is one settlement component, not the product.
A custodian	Custodians hold assets independently. Tenet co-signs vaults with FIs and LPs but holds no assets independently.
DTCC	DTCC tokenizes and clears traditional securities. Tenet settles the liquidity and delivery layer for tokenized assets on-chain. Complementary, not competitive.

2.2 The Four Engines

Tenet's internal architecture is composed of four engines. Each has a distinct responsibility. Together they handle the complete institutional trade lifecycle.

Engine	Responsibility	Functions
Orchestration Engine	The conductor	Receives FI intent, coordinates all other engines, writes every order state to the clearing ledger, manages the complete order lifecycle from intent to LP settlement.
Reservation Engine	The race-condition guard	Sub-component of Orchestration. Gate 1: locks FI vault balance before LP routing begins. Gate 2: locks LP vault inventory after fill plan is confirmed. Releases all locks on completion or failure.
Execution Engine	The trading brain	Runs price discovery across LP vaults, calculates slippage, performs LP depth analysis, produces execution plan. Assigns single LP or constructs multi-LP split plan by best price and depth. Handles asset delivery.
Settlement Engine	The capital movement layer	Extracts settlement from FI vault simultaneously with delivery. Routes LP payout via CCTP, internal netting, or Everclear. Receives only FI Tenet address and settlement amount — derives vault co-sign key internally.

3. Core Entities

Tenet's system involves three categories of entity. Each has a defined role, a defined vault structure, and a defined legal agreement with Tenet.

3.1 Financial Institutions (FIs)

Financial institutions are the demand side of Tenet's network. They hold digital assets or tokenized RWAs in custody and need to trade, swap, convert, or rebalance them efficiently without managing LP relationships, bridge risk, or cross-chain complexity. Each FI has the following identity structure within Tenet:

```
Tenet_address:       GoldmanSachs_x800        // Unique FI identifier
FI_identity:         Goldman Sachs Asset Mgmt  // Legal entity name
Settlement_vault:    GoldmanSettlementVault    // Co-signed with Tenet
Destination_wallets: [GoldmanWallet_SOL,
                      GoldmanWallet_ARB,
                      GoldmanWallet_ETH]       // Per-chain delivery
addresses
Compliance_data:     { jurisdiction, KYB, AML }
Signing_key:         <FI multisig key>
```

The Tenet_address is the master FI identifier. From it, Tenet derives the FI's identity, compliance status, settlement vault location, fee history, and complete order history. Every order ever placed by that FI is queryable by Tenet_address.

3.2 Liquidity Providers (LPs)

Liquidity providers are the supply side of Tenet's network. They are institutional market makers with deep inventory across multiple chains who want access to institutional order flow without managing FI relationships directly. Like FIs, every LP has a master Tenet address that serves as their identity within the system:

```
Tenet_address:    Wintermute_x400          // Unique LP identifier
LP_identity:      Wintermute Trading Ltd   // Legal entity name
Liquidity_vaults: [Wintermute_SOL,
                   Wintermute_ARB,
                   Wintermute_ETH]         // Per-chain vault addresses
Settlement_chain: SOL                      // LP preferred settlement
chain
Compliance_data:  { jurisdiction, KYB, AML }
Signing_key:      <LP multisig key>
Routing_tier:     1                        // Priority routing tier (1-3)
```

The LP Tenet address is the master LP identifier. From it, Tenet derives the LP's identity, compliance status, all vault addresses across all chains, settlement preferences, routing tier, fill history, and complete settlement history. Every order ever filled by that LP is queryable by Tenet_address. This makes the system fully bilateral — every order is traceable from both the FI side and the LP side, and Tenet holds the complete record for both.

3.3 Tenet

Tenet is the orchestration layer. It receives intents from FIs, sources and routes liquidity from LPs, executes delivery with orchestrated settlement extraction, routes settlement to LPs via the appropriate cross-chain protocol, maintains the clearing ledger, generates reporting for both FIs and LPs, and collects a tiered execution fee per order. Tenet does not custody funds independently. All assets live in co-signed vaults.

4. Vault Architecture

Tenet's asset flow model is built on four structurally distinct vault types. Each vault type has a specific purpose, a specific co-signing arrangement, and a specific legal basis. No assets flow outside this four-vault structure.

4.1 FI Settlement Vault

Property	Detail
Co-signers	FI + Tenet (2-of-2 multisig)
Purpose	Holds FI settlement funds. Tenet extracts per order as execution completes.

Funding model	FI pre-funds vault once. Drawn against across multiple simultaneous orders — the one pool of cash.
Reservation	Full settlement amount (spend + fee) is reserved by the Reservation Engine the moment an intent is accepted — before LP routing begins. Race condition protection.
Extraction trigger	Simultaneous with asset delivery to FI destination wallet. Orchestrated atomicity.
Insufficient funds	Reservation fails immediately. Order rejected with INSUFFICIENT_FI_FUNDS. No LP interaction initiated.
Vault count	One per FI. Multiple simultaneous orders draw from the same vault.
Custodian	Fireblocks or Copper (institutional-grade, insured)

The FI settlement vault is the structural innovation that enables the 'one pool of cash' model. Instead of requiring FIs to fund each order individually, the vault acts as a standing liquidity reserve. The Reservation Engine locks the required balance the moment an intent is accepted, ensuring no two concurrent orders can count the same funds.

4.2 LP Liquidity Vault

Property	Detail
Co-signers	LP + Tenet (2-of-2 multisig)
Purpose	Holds LP inventory. Tenet initiates transfer on order match via co-signing authority.
Funding model	LP pre-positions assets per chain. No per-order pre-lock required.
Reservation	Inventory reserved by the Reservation Engine after fill plan is confirmed. Available balance reduced immediately for concurrent order routing.
Vault count	One per LP per chain. An LP on 12 chains has 12 vaults.
Naming convention	LPName_Chain (e.g., Wintermute_SOL, Galaxy_ARB) — encodes LP identity and execution chain in one field.
Single LP orders	Assets transfer directly from LP vault to FI destination wallet. No aggregation step.
Multi-LP orders	Assets transfer from LP vault to Aggregation Vault first, then single delivery to FI.
Custodian	Fireblocks or Copper (LP selects, Tenet co-signs)

4.3 Aggregation Vault

Property	Detail
Owner	Tenet (sole custody)

Purpose	Temporary clearing account for multi-LP orders only. Combines assets from multiple LP vaults into the full order amount before single delivery to FI.
Used when	Order is split across two or more LPs. Not used for single LP orders.
FI experience	FI always receives one clean transaction regardless of how many LPs contributed.
Lifecycle	Created per multi-LP order. Closed and zeroed on order completion.

4.4 Tenet Settlement Vault

Property	Detail
Owner	Tenet (sole custody)
Purpose	Receives extracted settlement. Routes to LPs via CCTP, netting, or Everclear.
Inflow	Settlement extracted from FI Settlement Vault simultaneously with asset delivery.
Outflow	LP payout minus Tenet fee, routed per LP's settlement method.
Fee accumulation	Tenet's tiered fee accumulates here. Over time becomes proprietary inventory.
Settlement methods	CCTP (USDC), internal netting (same-chain multi-LP), Everclear (other cross-chain)

5. The Clearing Ledger

The clearing ledger is Tenet's internal order management and audit system. Every order has a complete lifecycle record from intent submission to LP settlement confirmation. The ledger is the operational backbone of Tenet's compliance, audit, and dispute resolution capabilities. It is the Orchestration Engine's only interface — the engine writes; the ledger displays.

5.1 Order Record Structure

```
OrderID:             ORD_93847592           // Unique order identifier
asset:               ETH                    // Asset being traded
usdc_amount:         500,000 USDC           // How much USDC the FI is
spending
asset_qty:           150.68 ETH             // Derived: usdc_amount /
spot_price
price_total:         500,500 USDC           // usdc_amount + Tenet fee
(extracted from vault)
slip_tolerance:      50 bps                 // FI-specified. Default: 50
bps.
eff_slippage:        12 bps                 // Actual slippage achieved
across fills
destination_wallet:  GoldmanWallet_SOL      // FI delivery destination
```

```
LP_assigned:          Wintermute_SOL          // LP + chain (implicit)
Tenet_address:        GoldmanSachs_x800       // FI identifier (implies
vault + history)
deadline:             30s                     // Chain-derived. SOL=15s,
ARB=30s, ETH=90s
status:               LIVE                    // Current lifecycle state
timestamp:            2026-03-09T14:23:01Z    // Order creation time
execution_tx:         0xabc...def             // On-chain delivery tx hash
settlement_tx:        0x123...456             // On-chain settlement tx hash
fee_collected:        500 USDC                // Tenet 10bps fee
```

5.2 Field Derivation Logic

Three fields carry implicit information that drives all downstream operations:

- usdc_amount — the FI's stated intent. The Execution Engine derives asset_qty internally from the current spot price. FIs always think and transact in USDC, never in asset units.
- LP_assigned (e.g., Wintermute_SOL) — encodes LP identity, execution chain, and target vault. No separate chain field required.
- Tenet_address (e.g., GoldmanSachs_x800) — maps to FI identity, jurisdiction, compliance records, settlement vault address, and full order history.

5.3 Ledger Partitions

Partition	Contents
LIVE	Orders where FI and LP reservations are confirmed and delivery is being initiated.
DELIVERING	Orders where LP vault transfer is in progress and FI settlement is being extracted simultaneously.
AWAITING SETTLEMENT	Orders where FI delivery is complete but LP settlement payout is pending.
COMPLETED	Fully settled orders. Immutable historical record. Available for audit.
FAILED	Cancelled, timed out, or errored orders with reason codes attached.
RECONCILIATION	Unmatched vault deposits requiring manual or automated review.
FI RESERVATIONS	Live view of all FI vault reservations: vault balance, locked amount, available, order breakdown, time remaining on deadline.
LP RESERVATIONS	Live view of all LP vault reservations: per-vault summary with per-order breakdown.

5.4 Reservation Engine

The Reservation Engine is a sub-component of the Orchestration Engine that owns all vault reservation logic for both FI and LP vaults. It is never called directly — only Orchestration interacts with it.

Gate 1 — FI vault reservation. The Reservation Engine fires first, before LP routing begins. It locks the full settlement amount (usdc_amount + fee) against the FI's settlement vault. If the vault cannot cover the full amount, the order fails immediately with INSUFFICIENT_FI_FUNDS. No LP interaction is initiated.

Gate 2 — LP vault reservations. After the Execution Engine confirms the fill plan, the Reservation Engine posts reservations against every LP vault in the plan. If any LP vault reservation fails, all partial LP reservations roll back immediately, the FI reservation is released, and the order fails cleanly.

Both gates must pass before an order reaches LIVE state. All reservations are released on completion or failure. The ledger records every reservation event with timestamp, vault ID, amount, and order ID.

5.5 Chain-Aware Deadlines

The flat 60-second deadline from prior versions has been replaced with a chain-aware reservation window derived from the finality characteristics of each chain. The deadline replaces and supersedes the earlier fixed-deadline model.

Chain	Reservation Window
SOL (Solana)	15 seconds — 400ms blocks, near-instant finality
ARB (Arbitrum)	30 seconds — 250ms Nitro blocks
BASE	30 seconds — OP stack, 2s blocks
POLYGON	45 seconds — 2s blocks, PBFT finality
ETH (Ethereum mainnet)	90 seconds — 12s blocks, 7-confirmation standard

For cross-chain orders (delivery chain differs from settlement chain), the deadline is max(delivChain, settlChain) — the more lenient rule, giving the slower chain its full window. FIs may extend the deadline beyond the chain floor by specifying an optional deadline field in the intent, but cannot shorten it below the floor. The engine enforces the minimum regardless of what the FI specifies.

5.6 Concurrent Order Isolation

When multiple FI orders route to the same LP vault simultaneously, isolation is achieved through the clearing ledger rather than on-chain tagging. The engine maintains a real-time map of expected LP vault inflows per active order. Asset deposits are matched to the oldest unmatched pending order for that vault, asset, and amount (FIFO). LPs are never required to embed order references in their transfers. The intelligence lives entirely within Tenet's engine.

6. Reporting

Tenet generates structured reports for both FIs and LPs across five time periods: end of day, end of week, end of month, end of quarter, and end of year. Because Tenet holds the complete bilateral order record — every intent, every execution, every settlement, every fee — it becomes the single source of truth for institutional digital asset activity on both sides of every trade.

This is not a secondary feature. Reporting is part of Tenet's core value proposition. FIs and LPs both carry significant back-office workload today — reconciliation, treasury management, audit trail maintenance, regulatory reporting, fee verification. Tenet's reporting layer absorbs a meaningful portion of that workload automatically.

6.1 FI Reports

Report	Contents
End of Day	All orders placed, assets delivered, settlement extracted, fees paid, vault balance remaining. Per-order breakdown with execution timestamps and transaction hashes.
End of Week	Weekly volume summary, average fill price vs market benchmark, fee total, chain breakdown, failed order log with reason codes.
End of Month	Monthly P&L attribution by asset and chain, cumulative fees, LP distribution summary, vault utilisation rate, settlement timeline averages.
End of Quarter	Quarterly trading summary, fee tier analysis, chain exposure breakdown, reconciliation log, compliance-ready audit trail.
End of Year	Full annual trading record, total fees paid, asset flow by chain, benchmark comparison, regulatory-grade audit export.

6.2 LP Reports

Report	Contents
End of Day	All orders filled, assets delivered from each vault, settlement received per chain, fees earned, vault balance remaining per chain.
End of Week	Weekly fill volume, fill rate by asset and chain, settlement timing averages, routing tier performance, inventory utilisation.
End of Month	Monthly revenue summary, vault turnover by chain, settlement method breakdown (CCTP vs netting vs Everclear), fill history vs inventory depth.
End of Quarter	Quarterly performance by chain and asset, routing tier review, settlement guarantee compliance record, reconciliation log.
End of Year	Full annual fill record, total revenue by asset and chain, inventory efficiency analysis, regulatory-grade settlement audit export.

6.3 Tenet as Source of Truth

Because every order flows through Tenet and every state transition is recorded in the clearing ledger, Tenet holds a complete and independently verifiable record of all activity. Neither FIs nor LPs need to cross-reference their own internal systems against each other to reconcile. Tenet's

report is the authoritative record. This creates a durable operational dependency — once FIs and LPs build their treasury workflows around Tenet's output, switching away means rebuilding that entire back-office infrastructure elsewhere.

7. Order Execution Lifecycle

A Tenet order moves through four primary states and two exception states. Every state transition is logged in the clearing ledger with a timestamp and relevant transaction data.

7.1 Intent Structure

Every FI interaction with Tenet begins with a structured intent. The intent schema is deliberately minimal — Tenet derives everything it can from the FI profile and internal engine logic, requiring the FI to specify only what is genuinely decision-relevant per order.

7.1.1 Standard Intent Fields

```
intent:           buy | sell | swap | convert
asset:            ETH | SOL | USDC | BTC | etc
amount:           500000          // Always USDC. How much to spend
(buy/swap)
                                  // or USDC notional of asset being sold
(sell).
                                  // Asset quantity is derived by the
engine.
settlement_asset: USDC | ETH | etc
delivery_chain:   SOL | ARB | ETH | BASE | POLYGON | etc
settlement_chain: ETH | SOL | etc   // buy and sell only
source_chain:     ETH | SOL | etc   // swap only
slip_tolerance:   50              // Optional. Basis points. Default: 50.
                                  // FI-specified. Tenet enforces it.
deadline:         120             // Optional. Seconds. FI extension only.
                                  // Cannot shorten below chain floor.
                                  // Engine derives floor from chain pair.
```

The amount field is always denominated in USDC — it represents how much the FI wants to spend (for buy and swap intents) or the USDC notional value of the asset being sold (for sell intents). Asset quantity is never specified by the FI. The Execution Engine derives it internally from the current spot price at the moment the intent is processed.

7.1.2 What Tenet Derives Automatically

Field	Source
Destination wallet	FI profile — registered per-chain wallet address. Never specified per intent.
Asset quantity	Execution Engine — derived from usdc_amount / spot_price at time of processing.
Slip tolerance	FI-specified (optional, default 50bps). The Execution Engine filters out any LP whose fill price exceeds the tolerance. If not specified, 50bps is used.

Deadline (floor)	Reservation Engine — derived from chain finality characteristics. SOL: 15s, ARB/BASE: 30s, POLYGON: 45s, ETH: 90s. Cross-chain: max(delivChain, settlChain).
LP assignment	Execution Engine — best price, best depth, single or multi-LP split.

7.1.3 Schedule Intent — Mode 1 (Inline)

Any standard intent can be scheduled for future execution by appending .schedule to the command. The CLI immediately prompts for date and time. The intent is stored in the clearing ledger with a SCHEDULED status and executed automatically at the specified time.

```
buy ETH 500000 USDC ARB ETH .schedule
→ Date: 2026-03-20
→ Time: 09:00 UTC
```

7.1.4 Schedule Intent — Mode 2 (Batch)

The FI opens a scheduling session by typing schedule first. Tenet prompts for date, time, and failure behaviour. The FI then enters an unlimited series of intents. All intents in the batch execute in parallel at the scheduled time.

```
schedule
→ Date: 2026-03-20
→ Time: 09:00 UTC
→ On failure: continue | halt
→ Intent 1: buy ETH 500000 USDC ARB ETH
→ Intent 2: sell SOL 200000 USDC SOL ETH
→ Intent 3: swap BTC 100000 ETH SOL
→ done
```

7.1.5 Convert Intent

Convert is a fiat-to-USDC onramp facilitated via Circle. The FI specifies the fiat amount and currency. Tenet coordinates the Circle instruction and monitors for USDC arrival in the FI's settlement vault. The fiat leg is handled entirely by Circle — Tenet does not touch fiat. A flat coordination fee is charged by Tenet. Circle's conversion fee is passed through transparently.

```
intent:   convert
amount:   1000000
currency: USD | GBP | EUR | etc
```

7.1.6 Intent Fee Structure

Intent	Fee

buy / sell	Volume-tiered: 10bps (up to $10M) → 7bps ($10M–$100M) → 5bps ($100M–$250M) → Negotiated ($250M+)
swap	Volume-tiered: same tier structure as buy / sell
Tokenized RWAs (any intent)	0.5bps flat — all order sizes
schedule	No additional fee — underlying intent fee applies at execution
convert	Flat coordination fee (TBD) + Circle conversion fee passed through

7.2 Supported Intent Types

Intent	Description
buy	Spend a specified USDC amount to purchase an asset on a specified delivery chain. Asset quantity derived from spot price by the engine.
sell	Sell an asset with a specified USDC notional value. Proceeds settle into FI vault on settlement chain. Asset quantity derived from spot price.
swap	Exchange one asset for another across chains. USDC notional of the source asset specified. Source chain specifies where the source asset currently lives.
convert	Fiat-to-USDC onramp via Circle. USDC lands in FI settlement vault. No chain fields required.
schedule	Container intent. Wraps one (Mode 1) or multiple (Mode 2) standard intents for future parallel execution.

7.3 State: LIVE

Both FI and LP vault reservations are confirmed. The Execution Engine has locked a fill plan. The Orchestration Engine has written the order to the clearing ledger. Asset delivery and settlement extraction are about to fire simultaneously.

7.4 State: DELIVERING

Assets are transferring from LP vault(s) to the FI's destination wallet. Simultaneously, the Settlement Engine is extracting the full settlement amount from the FI's vault into Tenet's settlement vault. Delivery and extraction are orchestrated to occur in the same execution window — orchestrated atomicity. Neither leg completes without the other. There is no settlement gap.

For multi-LP orders, assets from each LP vault transfer into a temporary Aggregation Vault first. Once all LP legs are confirmed, a single delivery goes to the FI destination wallet. The FI always receives one clean transaction regardless of the number of LPs involved.

7.5 State: AWAITING_SETTLEMENT

The FI has received their asset. From the FI's perspective the transaction is complete in under 5 seconds. Settlement funds are in Tenet's settlement vault, queued for LP payout. Tenet deducts its tiered fee at this stage before routing LP settlement.

7.6 State: COMPLETED

LP settlement confirmed on their native chain. All reservations released. Full order record moved to the completed partition. Immutable. Available for FI reporting, LP verification, regulatory audit, and compliance review.

7.7 Exception: FAILED

Order could not execute. Reason codes: INSUFFICIENT_FI_FUNDS, LP_LIQUIDITY_UNAVAILABLE, SLIPPAGE_EXCEEDED, CHAIN_TIMEOUT, VAULT_CO-SIGN_FAILURE. All vault reservations released immediately. Full failure record preserved with reason code and detail.

7.8 Exception: RECONCILIATION

An asset deposit arrived in a vault that cannot be matched to an active order. Assets held in reconciliation partition until the discrepancy is resolved manually or automatically.

8. Multi-LP Split Logic

Large institutional orders frequently exceed the inventory depth of any single LP on a given chain. Tenet's engine handles this via slippage-aware split logic across multiple LPs.

8.1 Split Execution Example

```
FI Intent: Spend $50,000,000 USDC on ETH on Solana  (slip_tolerance:
50bps)

Engine derives: ~15,070 ETH at spot $3,318.50

LP Quotes (filtered to within 50bps):
  Wintermute_SOL:  available 5,776 ETH  @ 0.99981  (1.9bps)
  Galaxy_SOL:      available 5,130 ETH  @ 0.99989  (1.1bps)
  Cumberland_SOL:  available 4,094 ETH  @ 0.99995  (0.5bps)

Tenet Split Assignment:
  ORD_001  Wintermute_SOL   5,776 ETH   38.3%
  ORD_002  Galaxy_SOL       5,130 ETH   34.0%
  ORD_003  Cumberland_SOL   4,094 ETH   27.2%
  Residual routed to next available LP

  Eff. slippage: 14bps  (within 50bps tolerance  ✓)
```

8.2 Ledger Handling for Splits

Each LP leg receives its own order record linked to a parent intent ID. The parent intent is not marked COMPLETED until all child orders reach COMPLETED state. This ensures atomic completion from the FI's perspective even when multiple LPs are involved.

```
Parent Intent:  INT_20260309_001  status: LIVE -> COMPLETED
  Child ORD_001  Wintermute_SOL   5,776 ETH   status: COMPLETED
  Child ORD_002  Galaxy_SOL       5,130 ETH   status: COMPLETED
  Child ORD_003  Cumberland_SOL   4,094 ETH   status: COMPLETED
```

9. Settlement Routing

After FI delivery is confirmed and settlement funds are in Tenet's settlement vault, Tenet routes LP payouts via one of three methods depending on asset type and chain.

9.1 CCTP — Circle Cross-Chain Transfer Protocol

Used for all USDC settlement across chains. CCTP burns USDC on the source chain and mints native USDC on the destination chain. No wrapped token risk. No bridge counterparty. No liquidity pool dependency.

Property	Value
Asset scope	USDC only
Mechanism	Burn on source chain, mint on destination chain
Chains supported	Ethereum, Arbitrum, Base, Optimism, Avalanche, Solana, Polygon
Settlement time	2-5 minutes (v1). Sub-minute targeted in CCTP v2.
Bridge risk	None. Native Circle protocol.

9.2 Internal Netting

When multiple LP settlement obligations exist on the same chain, Tenet nets them internally before executing any on-chain transfers. This minimises gas costs and reduces settlement transaction volume. Internal netting is applied wherever possible before falling back to CCTP or Everclear.

9.3 Everclear

Used for non-USDC assets requiring cross-chain settlement where internal netting is insufficient. Tenet transfers to Everclear's clearing layer, which handles netting and delivers final settlement to the LP's native chain. Bridge fees are passed through transparently to the FI, included in price_total at quote time.

10. Failure Handling

Every failure mode in Tenet's execution lifecycle has a defined response. Nothing hangs indefinitely. Every exception produces a reason code, a ledger entry, and a defined resolution path. All vault reservations are released on any failure path.

Failure	Trigger	Response
INSUFFICIENT_FI_FUNDS	FI vault reservation fails at Gate 1	Order rejected immediately. FI signalled with required top-up. No LP interaction. Reservation never posted.
VAULT_PENDING_CUSTODY	FI vault not yet connected to custody provider	Order rejected. FI notified to complete vault portal connection.
SLIPPAGE_EXCEEDED	No LP within FI's slip_tolerance on the requested chain	Order rejected. FI notified with available LP pricing. FI may resubmit with wider tolerance.
LP_LIQUIDITY_UNAVAILABLE	No LP combination fills the order at any slippage	Order paused. FI notified. Rerouted if alternate LP available. Otherwise cancelled. FI reservation released.
LP_PARTIAL_FILL	LP vault balance less than assigned amount post-reservation	Remaining quantity rerouted to next best LP. If impossible, FI offered partial fill.
CHAIN_TIMEOUT	Reservation deadline exceeded for chain combination	Reservation Engine flags STALE. Orchestration cancels order. All vault reservations released. Assets returned.
SETTLEMENT_ROUTING_FAILURE	CCTP or Everclear unavailable at settlement	Settlement held in Tenet vault. LP notified. Awaiting settlement until routing resumes.
VAULT_CO-SIGN_FAILURE	Custody API unavailable for co-signing	Order paused. Retry with exponential backoff. Manual intervention path available.
RECONCILIATION	Unmatched vault deposit	Assets held in reconciliation. Automated matching attempted. Manual review if unresolved after 1 hour.

11. Legal Framework & Agreements

Tenet's legal structure is designed to be asset-light while maintaining clear contractual responsibility for delivery and settlement. Two core agreement types govern all counterparty relationships.

11.1 FI Master Agreement

Every financial institution onboarding to Tenet signs a Master Agreement covering the following terms:

11.1.1 Vault Co-Signing Authority

The FI grants Tenet co-signing authority over the FI Settlement Vault, scoped exclusively to settlement extraction for completed orders and reservation management by the Reservation Engine. Tenet may not extract from the vault for any other purpose. The FI retains the right to withdraw funds at any time subject to pending order obligations and active reservations.

11.1.2 Intent Submission & Binding Execution

A submitted intent becomes binding upon Tenet's confirmation of sufficient vault balance (Gate 1 reservation) and LP availability (Gate 2 reservation). Cancellation is permitted only before LP vault transfer is initiated.

11.1.3 Tenet's Delivery Obligation

Tenet warrants that upon order acceptance, the specified asset will be delivered to the FI's designated destination wallet within the execution deadline, or the order will be cancelled and any extracted settlement returned. This is Tenet's primary contractual liability to FIs.

11.1.4 Fee Structure

Tenet charges 10 basis points of the USDC amount on standard orders (volume-tiered for larger orders), extracted simultaneously with settlement. The fee is disclosed in the price_total field of every order quote. No additional fees are charged without prior notice and FI consent.

11.1.5 Data & Audit Rights

The FI has the right to request a complete audit trail of any order, including clearing ledger records, on-chain transaction hashes, LP identities, and settlement confirmations. Tenet retains order records for a minimum of seven years.

11.1.6 KYB & AML Compliance

FI onboarding requires Know Your Business verification and AML screening. Tenet reserves the right to suspend FI access if compliance status changes or suspicious activity is detected.

11.1.7 Liability Cap

Tenet's aggregate liability to any FI is capped at the total fees paid by that FI in the preceding 12 months. Tenet is not liable for chain-level failures, custody provider outages, or regulatory actions beyond its control. Tenet is liable for losses arising directly from breach of its delivery obligation.

11.2 LP Liquidity Access Agreement

Every liquidity provider onboarding to Tenet signs a Liquidity Access Agreement covering the following terms:

11.2.1 Vault Co-Signing Authority

The LP grants Tenet co-signing authority over each LP Liquidity Vault, scoped exclusively to order execution and Reservation Engine inventory management. Tenet may not extract LP assets for any other purpose. The LP retains the right to withdraw assets at any time subject to pending order obligations and active reservations.

11.2.2 Inventory Visibility

By maintaining funded LP vaults, the LP grants Tenet visibility into available inventory for price discovery and routing. Tenet will not share LP inventory data with FIs or third parties. LP inventory data is used exclusively for internal routing calculations.

11.2.3 Priority Routing Tiers

LPs are assigned to routing priority tiers based on inventory depth, tenure, and fill history. Higher-tier LPs receive first routing consideration for new orders. Priority tier assignments are reviewed quarterly.

11.2.4 Settlement Guarantee

Tenet guarantees LP settlement within the timeframe specified per settlement method: CCTP within 10 minutes of order completion, Everclear within the protocol's confirmed window, internal netting within 1 business day. If delayed, Tenet pays interest at SOFR + 2% per annum.

11.2.5 Native Chain Settlement

LPs will always receive settlement on their native chain as specified in their agreement. Tenet will not redirect settlement to an alternative chain without LP consent. This is a contractual guarantee, not a best-efforts commitment.

11.2.6 No FI Exposure

LP identity will not be disclosed to FIs. LPs will not be required to interact with FIs for any purpose. Tenet is the sole counterparty the LP ever sees.

11.2.7 Minimum Inventory Commitment

LPs agree to maintain a minimum inventory level per vault as specified in the agreement schedules. Failure to maintain minimum inventory may result in downgrade of priority routing tier with 30 days notice.

11.3 Non-Binding Letter of Intent — Design Partner Phase

During the design partner phase, Tenet enters non-binding Letters of Intent with both FIs and LPs establishing mutual intent to explore a design partner relationship, including:

- Participation in product feedback sessions and architecture review
- Non-disclosure of Tenet's proprietary architecture and clearing ledger design
- Right of first access to Tenet's testnet when available
- Priority consideration for production onboarding upon launch

The LOI converts to a binding Master Agreement or Liquidity Access Agreement upon Tenet's production launch and formal counterparty onboarding completion.

12. Technical Stack

Layer	Technology & Rationale
Intent Layer	gRPC — Sub-millisecond intent submission and real-time order state

	streaming. Built for institutional volume.
CLI Interface	Node.js + TypeScript — Working CLI for direct intent submission. USDC-denominated amounts. FI-specified slippage tolerance. Live CryptoCompare price feed integration.
Price Discovery	CryptoCompare API (primary) + CoinGecko (fallback) — 30-second cache, live pricing, multi-asset.
Custody / Vaults	Fireblocks + Copper — Institutional-grade co-signed vaults with MPC key management and insurance. VaultProvider abstract interface — custody layer connects without modifying engine logic.
Reservation Engine	Sub-component of Orchestration Engine — owns all FI and LP vault reservation logic. Chain-aware deadlines. Dual-gate race condition protection.
USDC Settlement	Circle CCTP — Native cross-chain USDC transfer. No bridge risk.
Non-USDC Settlement	Everclear — Cross-chain clearing and settlement for non-USDC assets.
Clearing Ledger	Proprietary internal ledger — Real-time lifecycle tracking, sheet-based partition display, FIFO concurrency resolution, full audit log. Engine's only interface.
Chains Supported	Ethereum, Solana, Arbitrum, Base, Polygon (extensible via LP vault expansion)
Language	TypeScript (Node.js) — Strict mode, type-safe intent and order data structures throughout.

13. Competitive Landscape

The institutional digital asset market has several participants who address parts of the liquidity and settlement problem. None address the full stack that Tenet orchestrates.

Player	Model	What They Miss
Talos	Trading infrastructure	Explicitly disclaims intermediary role. FIs and LPs negotiate directly. No clearing, no settlement, no atomic delivery.
FalconX	Principal dealer	Takes the other side of trades from its own balance sheet. Balance sheet risk. No neutral orchestration. No cross-chain settlement.
Hidden Road / Ripple	Prime broker	Acquired by Ripple in 2025. Now vertically integrated into XRP Ledger. Chain dependency most FIs will not accept.
DTCC	Tokenization	Solving asset issuance, not settlement routing. No cross-chain delivery, no LP network, no real-time orchestration.
Tenet Systems	Orchestration layer	N/A. Tenet is the missing layer. FIs and LPs never interact. Any chain. Asset-light. Volume-tiered pricing. Reservation Engine. Chain-aware deadlines.

The key insight: FIs only care about getting their asset quickly at the best price. LPs only care about order flow and reliable settlement. Neither wants to manage a relationship with the other. Tenet is the only system built from first principles around this insight.

14. Market Opportunity

Signal	Detail
Tokenized RWA market	$30 trillion projected by 2034 (Standard Chartered). Every tokenized asset requires a settlement layer.
Digital asset daily volume	$3 trillion+ globally. At 10bps, $1B daily volume generates $365M annually.
DTCC SEC approval	December 2025: DTCC approved to tokenize U.S. Treasuries on-chain. Issuance layer arriving. Settlement layer needed.
Hidden Road acquisition	Ripple acquired Hidden Road for $1.25B in 2025. Institutional settlement infrastructure commands premium valuations.
CCTP and Everclear	Cross-chain settlement protocols are production-ready today. Technical prerequisites for Tenet's routing exist.
Institutional adoption	ETFs, pension funds, sovereign wealth funds actively entering digital assets. Infrastructure demand is structural.

14.1 Fee Structure

Order Size / Asset Type	Tenet Fee
Up to $10M	10 basis points
$10M — $100M	7 basis points
$100M — $250M	5 basis points
$250M+	Negotiated
Tokenized RWAs (all sizes)	0.5 basis points flat

14.2 Revenue Projections

Daily Volume	Blended Annual Revenue	Stage
$100M	~$25M	Year 2 — design partner pilots
$1B	~$200M	Series A trajectory
$10B	~$2B	Scale
$500B	$100B+	15-20 year vision — institutional rails

The $500B daily volume figure is not speculative. SWIFT processes $5 trillion in daily messages. DTCC settles $2.15 trillion daily. As tokenized RWAs and digital assets converge with traditional finance, the institutional settlement layer for that combined market will process volumes in this range. Tenet is positioned at that intersection.

15. The Moat

Tenet's competitive advantages compound over time through four distinct mechanisms.

15.1 The Clearing Ledger as a Proprietary Data Asset

Every order executed through Tenet builds an immutable record in the clearing ledger. Over time this becomes an extraordinarily valuable dataset — a complete record of institutional digital asset flow, pricing, LP depth, and settlement patterns across all chains. This data cannot be replicated by any competitor who enters later. The ledger is a durable moat built one order at a time.

15.2 Network Effects on Both Sides

Every FI onboarded makes Tenet more valuable to LPs because it increases order flow. Every LP onboarded makes Tenet more valuable to FIs because it improves price discovery and fill rates. Both sides compound together. Switching away from Tenet means losing access to the other side's network — a cost that increases with every new participant.

15.3 LP Lock-In via Settlement Certainty

Native chain settlement is a contractual guarantee. LPs build treasury management workflows around this certainty. Fireblocks vault integrations, reconciliation systems, and treasury dashboards become embedded dependencies on Tenet's settlement reliability. Switching has a real and growing operational cost.

15.4 Proprietary Inventory

Tenet collects 10 basis points on every order. Over time, accumulated fees build a capital base that Tenet can deploy as proprietary inventory. Tenet then becomes both orchestrator and liquidity provider on high-volume orders, capturing the full spread rather than just the routing fee. This is the long-term margin expansion path.

16. Conclusion

The institutional digital asset market is at an inflection point. Tokenization of real-world assets is beginning in earnest. DTCC has regulatory approval to bring U.S. Treasuries on-chain. Sovereign wealth funds, pension funds, and asset managers are actively allocating. The infrastructure to support this activity — the clearing, settlement, and liquidity orchestration layer — does not yet exist at institutional grade.

Tenet Systems builds that infrastructure.

One intent. Any asset. Any chain. USDC-denominated. Slippage controlled by the FI. Delivered to the FI in under five seconds. Settled to the LP in the background. Race conditions eliminated

by the Reservation Engine before a single LP is contacted. Deadlines derived from chain finality, not a hardcoded clock. Every order recorded in a clearing ledger that becomes more valuable with every trade.

No counterparty exposure for FIs. No FI relationship management for LPs. Volume-tiered pricing that scales with the institutions it serves. Flat 0.5bps for tokenized RWAs — the fastest growing asset class in institutional finance.

The plumbing for institutional digital asset markets is being built now. Tenet is that plumbing.

DESIGN PARTNER ENQUIRIES *Tenet Systems is currently onboarding FI design partners and LP relationships. Contact us via tenet-systems.netlify.app or reach out to the founding team directly to discuss early access, the non-binding LOI process, and the testnet timeline.*

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